Lab 6 Fluorescence of Anthracene

1. Calculation for Quencher Solution

Given basic information of ethyl iodide,

$\rho = 1.950\ g/mL$ $M = 155.97\ g/mol$

(1) 0.050 M

$n = C \cdot V = 0.050M \cdot 0.025L = 0.00125 mol$

$m = nM = 0.00125 mol \cdot 155.97\ g/mol = 0.1949625g$

$V = \frac{m}{\rho} = \frac{0.1949625g}{1.950g/mol} \approx 0.0999809\ mL \approx 0.1\ mL$

(2) 0.100 M

$n = C \cdot V = 0.100M \cdot 0.025L = 0.0025\ mol$

$m = nM = 0.0025\ mol \cdot 155.97\ g/mol = 0.389925g$

$V = \frac{m}{\rho} = \frac{0.389925g}{1.950g/mol} \approx 0.199615\ mL \approx 0.2mL$

(3) 0.150 M

$n = C \cdot V = 0.150M \cdot 0.025L = 0.00375\ mol$

$m = nM = 0.00375 mol \cdot 155.97\ g/mol = 0.5848875g$

$V = \frac{m}{\rho} = \frac{0.5848875g}{1.950g/mol} \approx 0.299423\ mL \approx 0.3mL$

(4) 0.200 M

$n = C \cdot V = 0.200M \cdot 0.025L = 0.005 mol$

$m = nM = 0.005 mol \cdot 155.97 g/mol = 0.77985 g$

$V = \frac{m}{\rho} = \frac{0.77985g}{1.950g/mol} \approx 0.399923 mL \approx 0.4mL$

2. Calculation for Stern-Volmer Plot

* 340nm excitation, emission peaks: ① 378.46nm ; ② 399.53 nm ; ③ 423.03nm ; ④ 448.93nm

(1) 378.46nm

$F_0 = 385.147$ ① 0.05M: $F = 218.937$ $\frac{F_0}{F} = \frac{385.147}{218.937} \approx 1.759$

② 0.10M: $F = 142.858$ $\frac{F_0}{F} = \frac{385.147}{142.858} \approx 2.696$

③ 0.15M: $F = 105.302$ $\frac{F_0}{F} = \frac{385.147}{105.302} \approx 3.658$

④ 0.20M: $F = 82.423$ $\frac{F_0}{F} = \frac{385.147}{82.423} \approx 4.673$

(2) 399.530 nm

$F_0 = 428.588$ ① 0.05M: $F = 242.939$ $\frac{F_0}{F} = \frac{428.588}{242.939} \approx 1.764$

② 0.10M: $F = 158.142$ $\frac{F_0}{F} = \frac{428.588}{158.142} \approx 2.710$

③ 0.15M: $F = 116.796$ $\frac{F_0}{F} = \frac{428.588}{116.796} \approx 3.670$

④ 0.20M: $F = 91.481$ $\frac{F_0}{F} = \frac{428.588}{91.48} \approx 4.685$

(3) 423.030 nm

$F_0 = 188.755$ ① 0.05M: $F = 107.568$ $\frac{F_0}{F} = \frac{188.755}{107.568} \approx 1.755$

② 0.10M: $F = 70.318$ $\frac{F_0}{F} = \frac{188.755}{70.318} \approx 2.684$

③ 0.15M: $F = 51.849$ $\frac{F_0}{F} = \frac{188.755}{51.849} \approx 3.640$

④ 0.20M: $F = 40.517$ $\frac{F_0}{F} = \frac{188.755}{40.517} \approx 4.659$

(4) 448.93 nm

$F_0 = 39.568$ ① 0.05M: $F = 22.263$ $\frac{F_0}{F} = \frac{39.568}{22.263} \approx 1.777$

② 0.10M: $F = 14.552$ $\frac{F_0}{F} = \frac{39.568}{14.552} \approx 2.719$

③ 0.15M: $F = 10.865$ $\frac{F_0}{F} = \frac{39.568}{10.865} \approx 3.642$

④ 0.20M: $F = 8.408$ $\frac{F_0}{F} = \frac{39.568}{8.408} \approx 4.706$

3. Calculation for Ksv

From linear regressions:

① 378.460 nm: $k_1 = 18.488 \pm 0.549$; $b_1 = 0.908 \pm 0.067$ $(R^2 = 0.9974)$

② 399.530 nm: $k_2 = 18.551 \pm 0.532$; $b_2 = 0.911 \pm 0.065$ $(R^2 = 0.9975)$

③ 423.030 nm: $k_3 = 18.406 \pm 0.561$; $b_3 = 0.907 \pm 0.069$ $(R^2 = 0.9972)$

④ 448.930 nm: $k_4 = 18.553 \pm 0.562$; $b_4 = 0.914 \pm 0.069$ $(R^2 = 0.9973)$

$$\bar{k} = \frac{1}{n}\sum_{i=1}^{n} k_i = \frac{1}{4}\cdot(18.488+18.551+18.406+18.553) = 18.4995 \simeq 18.500$$

$$S_k = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(k_i-\bar{k})^2} = \sqrt{\frac{1}{3}[(18.488-18.4995)^2+(18.551-18.4995)^2+(18.406-18.4995)^2+(18.553-18.4995)^2]}$$

$$\approx 0.069$$

With 95% confidence interval:

$$\lambda_{K_{SV}} = \frac{t\cdot S}{\sqrt{N}} = \frac{3.182\cdot 0.069}{\sqrt{4}} \approx 0.110$$

So, $K_{SV} = 18.500 \pm 0.110\ M^{-1}$

4. Calculation for k_q

Given $\tau_0 = 4.83\text{ns} = 4.83\times10^{-9}\text{s}$

$$k_q = \frac{K_{SV}}{\tau_0} = \frac{18.500}{4.83\times10^{-9}} M^{-1}\cdot s^{-1} \approx 3.830\times10^{9} M^{-1}\cdot s^{-1}$$

$$\lambda_{kq} = \frac{\Delta K_{SV}}{\tau_0} = \frac{0.110}{4.83\times10^{-9}} M^{-1}s^{-1} \approx 0.023\times10^{9} M^{-1}s^{-1}$$

So, $k_q = (3.830 \pm 0.023)\times10^{9}\ M^{-1}\cdot s^{-1}$

5. Error Analysis

(1) Comparison with CPL

$K_{SV}^{lit} = 18.800 M^{-1}$

$$\text{error } K_{SV} = \frac{|18.500-18.800|}{18.800}\times100\% \approx 1.596\%$$

$$k_q^{lit} = \frac{18.800}{4.83\times10^{-9}} M^{-1}\cdot s^{-1} \approx 3.892\times10^{9} M^{-1}\cdot s^{-1}$$

$$\text{error } k_q = \frac{|3.830-3.892|}{3.892}\times100\% \approx 1.593\%$$

(2) Comparison with JACS

$K_{SV}^{lit} = (4.83\times10^{-9})\cdot(3.950\times10^{9})\, M^{-1}\cdot s^{-1} \approx 19.100\, M^{-1}$

error $K_{SV} = \frac{|18.500-19.100|}{19.100}\times100\% \approx 3.141\%$

$k_q^{lit} = 3.950\times10^{9}\, M^{-1}s^{-1}$

error $k_q = \frac{|3.830-3.950|}{3.950}\times100\% \approx 3.038\%$

6. Emission Spectrum Comparison

$\Delta\lambda_1 = (378.46-378.00)\,nm = 0.46\,nm$

$\Delta\lambda_2 = (399.53-399.53)\,nm = 0$

$\Delta\lambda_3 = (423.03-423.48)\,nm = -0.45\,nm$

$\Delta\lambda_4 = (39.568-33.457)\,nm = 0.610\,nm$

ratio $I_1 = \frac{234.751}{385.147} \approx 0.610$

ratio $I_2 = \frac{259.919}{428.588} \approx 0.607$

ratio $I_3 = \frac{120.237}{188.755} \approx 0.637$

ratio $I_4 = \frac{33.457}{39.568} \approx 0.846$

7. Excitation-emission Calculation $v = \frac{10^7}{\lambda}$

*Excitation: $v_1 = \frac{10^7}{376.000} \approx 26595.745\,cm^{-1}$ $v_4 = \frac{10^7}{324.000} \approx 30864.198\,cm^{-1}$

$v_2 = \frac{10^7}{356.460} \approx 28053.639\,cm^{-1}$ $v_5 = \frac{10^7}{309.530} \approx 32307.047\,cm^{-1}$

$v_3 = \frac{10^7}{339.530} \approx 29452.478\,cm^{-1}$

$$\Delta V_1 = (32307.047 - 30864.198)\,cm^{-1} = 1442.849\,cm^{-1}$$

$$\Delta V_2 = (30864.198 - 29452.478)\,cm^{-1} = 1411.720\,cm^{-1}$$

$$\Delta V_3 = (29452.478 - 28053.639)\,cm^{-1} = 1398.839\,cm^{-1}$$

$$\Delta V_4 = (28053.639 - 26595.745)\,cm^{-1} = 1457.894\,cm^{-1}$$

$$\Delta \overline{V} = \frac{1}{4}\cdot(\Delta V_1 + \Delta V_2 + \Delta V_3 + \Delta V_4) \approx 1427.826\,cm^{-1}$$

$$S_{\Delta \overline{V}} = \sqrt{\frac{1}{3}\sum_{i=1}^{4}(\Delta V_i - \Delta \overline{V})^2} \approx 27.257\,cm^{-1}$$

对 Emission:

$$V_1 = \frac{10^7}{378.460} \approx 26422.871\,cm^{-1} \qquad V_3 = \frac{10^7}{423.030} \approx 23638.985\,cm^{-1}$$

$$V_2 = \frac{10^7}{399.530} \approx 25029.410\,cm^{-1} \qquad V_4 = \frac{10^7}{448.930} \approx 22275.191\,cm^{-1}$$

$$\Delta V_1 = (26422.871 - 25029.410)\,cm = 1393.461\,cm^{-1}$$

$$\Delta V_2 = (25029.410 - 23638.985)\,cm = 1390.425\,cm^{-1}$$

$$\Delta V_3 = (23638.985 - 22275.191)\,cm = 1363.794\,cm^{-1}$$

$$\Delta \overline{V} = \frac{1}{4}\cdot(\Delta V_1 + \Delta V_2 + \Delta V_3 + \Delta V_4) \approx 1382.560\,cm^{-1}$$

$$S_{\Delta \overline{V}} = \sqrt{\frac{1}{3}\sum_{i=1}^{4}(\Delta V_i - \Delta \overline{V})^2} \approx 16.317\,cm^{-1}$$